CW1 Individual Report

3D Interactive Media Development

Student name	Tabitha Kajendran
Student ID	w2177362
Group	Group 7
Individual contribution	Design

The aim of this group project was to create a 3D interactive media product inspired by Disney's Treasure Planet. The project involved developing a 3D asset based on the film's combination of traditional sailing ships and science-fiction elements, which would form part of the group's final Unity project. My individual responsibility within Group 7 was Ship Design and Customisation. Another group member was responsible for sourcing and importing the original ship model, while my role focused on developing its visual appearance using Mecabricks. I was responsible for making design decisions and customising the existing model to better reflect our chosen theme. My main contributions included introducing a royal blue and gold colour scheme, changing the front deck from black to blue and gold, and removing the black skull decoration from one of the sails to create a plain white finish. I also reviewed the model from different angles to assess the overall appearance. This report discusses the inspiration behind my design, the development process, the decisions I made, the challenges I experienced and the skills I gained through contributing to the group's project.

My design concept was influenced by the visual style and adventurous atmosphere of *Treasure Planet*. The film combines traditional sailing ships with futuristic science-fiction features, creating a distinctive setting that mixes familiar historical elements with imaginative designs. I wanted my ship to reflect this combination while maintaining a clear and consistent visual identity. Rather than attempting to copy an existing ship from the film, I developed my own interpretation of the theme through colour and decorative choices. I decided that royal blue, gold and white would form the main colour palette. Royal blue was selected as the dominant colour because I felt it would give the ship a strong appearance and help distinguish it from a traditional wooden sailing ship. Gold was chosen as a complementary colour to introduce contrast and make the design more decorative. Together, these colours helped establish the appearance I wanted without introducing too many competing colours. I also decided to keep the sails white. This was partly influenced by my preference for a simpler design, as I wanted the sails to contrast with the more colourful sections of the ship. I chose to remove the black skull decoration from one sail because I felt it did not suit the appearance I was trying to achieve. My overall intention was to

create a ship that looked adventurous and visually interesting while remaining recognisable as a traditional sailing vessel. I wanted the customisation to support the group's Treasure Planet theme rather than completely change the original model.

The design development process began after another group member sourced and imported the original ship model. I then opened the existing model in Mecabricks and examined its structure and appearance before making any changes. This allowed me to identify the sections that needed customisation and consider how the new colour scheme could be applied. My first main task was changing the original colour scheme. I introduced royal blue and gold across the ship, paying attention to how the colours appeared alongside the existing structure and decorative details. I wanted the colours to be distributed in a way that made the different sections look connected rather than making individual areas appear separate. One area I focused on was the front deck, which was originally black. I changed this section to blue and gold because the black colouring did not fit the direction of my design. This adjustment helped the front section match the rest of the ship and made the colour scheme more consistent. I then moved on to the sails. I wanted to retain a plain white appearance, so I removed the black skull decoration from one of the sails. This was an important part of the customisation because it allowed me to achieve the simpler appearance I had planned. During the editing process, I encountered difficulties selecting individual bricks because selecting a particular part initially caused the entire ship to become selected. This made certain adjustments more difficult and meant I had to be careful when working on different sections of the model. It also affected the amount of time needed to complete the customisation. After making the changes, I reviewed the ship from different angles to check how the colours and modified sections worked together. This helped me assess the overall appearance rather than judging the design based on one viewing angle.



Figure 1: Original ship model before design



Figure 2: Customisation of the ship's colour scheme and front deck

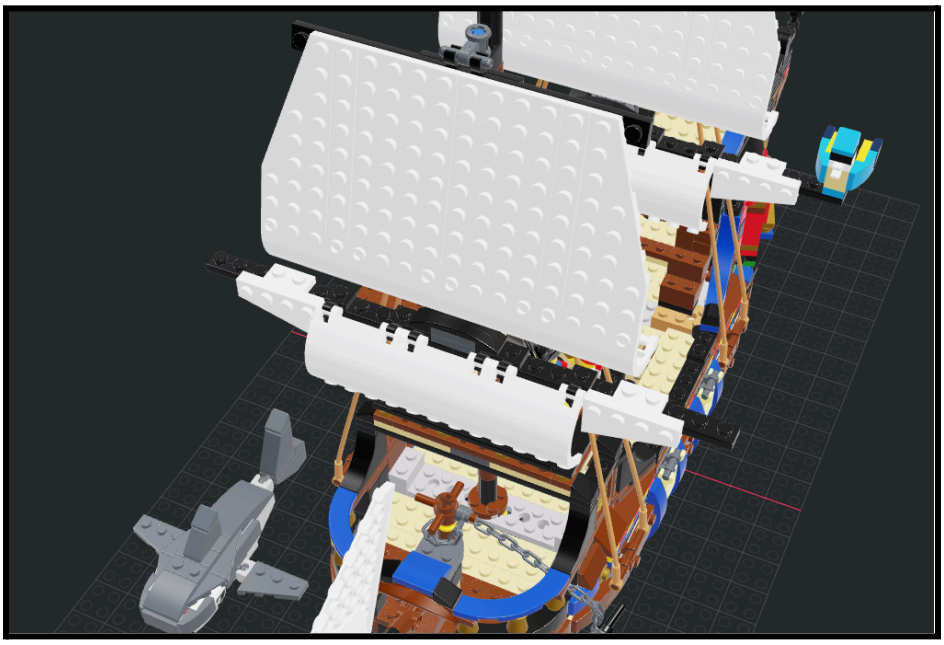


Figure 3: White sail after removing the black skull decoration

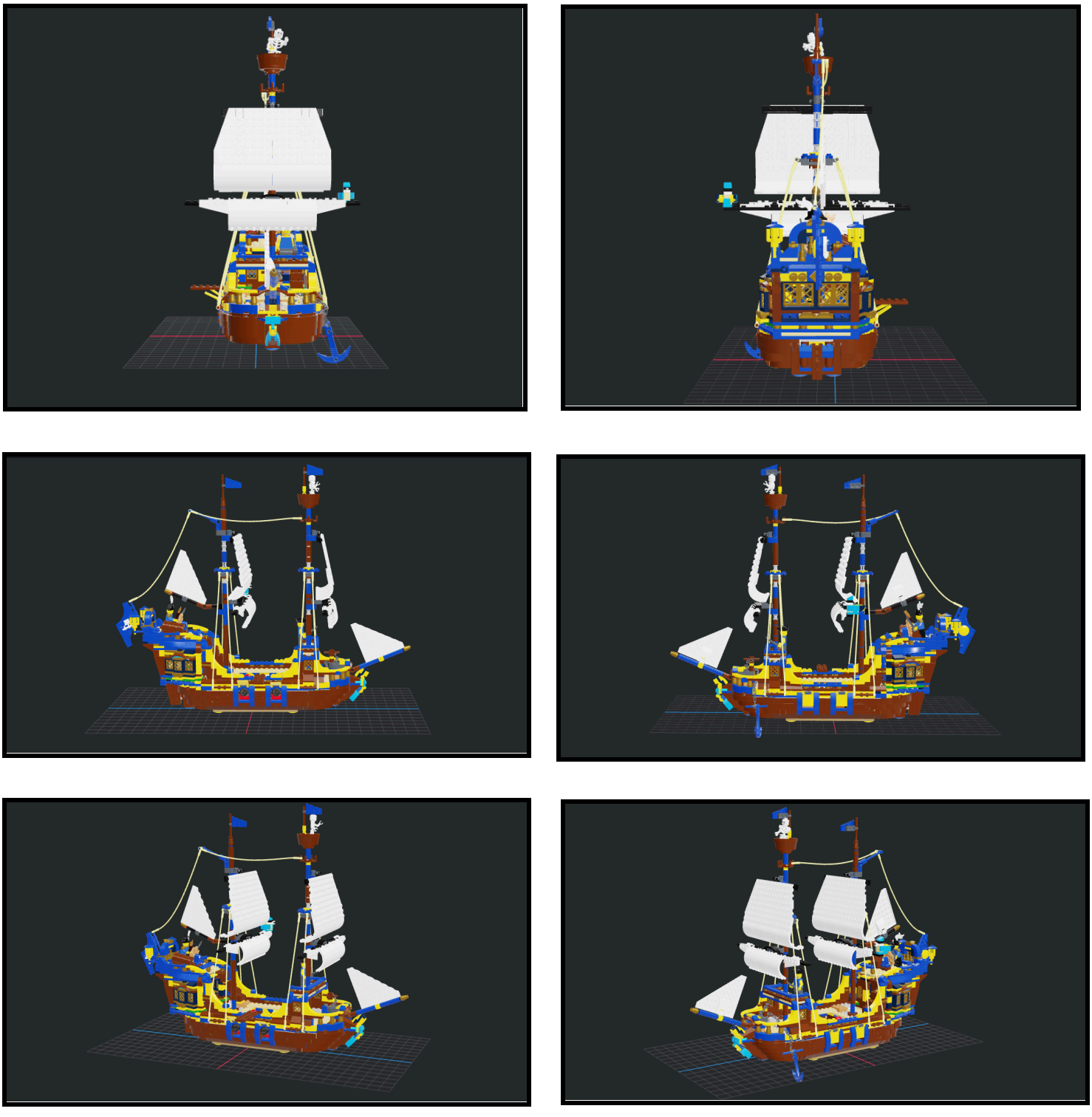
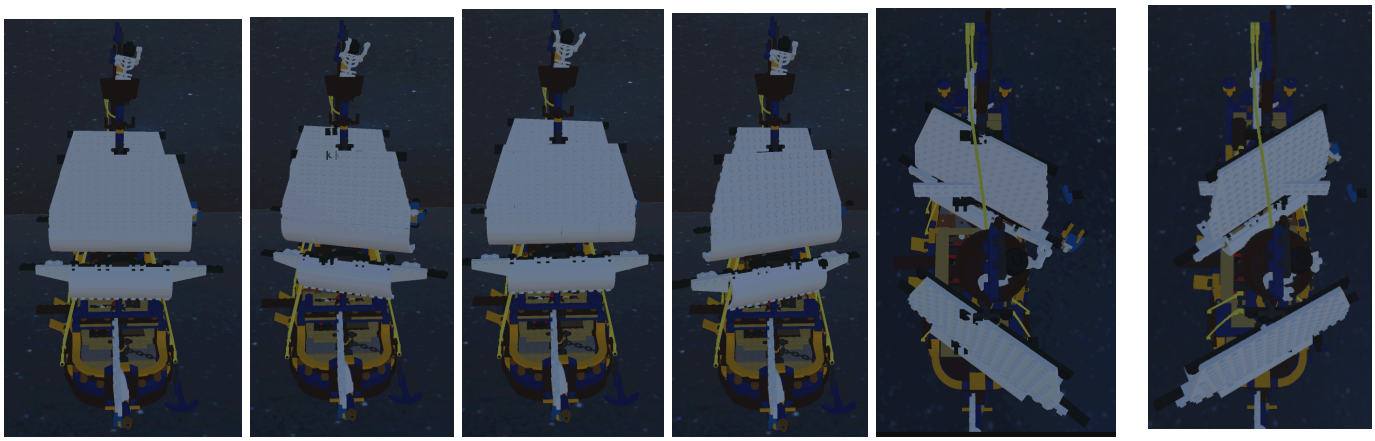


Figure 4: Completed ship from different viewing angles

After completing the customisation, I evaluated how successfully the changes achieved my original design concept. One of my main considerations was whether the different colours worked together across the entire ship. I was particularly satisfied with the distribution of royal blue and gold. The colours provided enough contrast to make different sections noticeable while maintaining a consistent appearance. Changing the front deck from black was also effective because it removed a colour that did not fit the rest of the design. The front section now followed the same visual direction as the remaining parts of the ship. The white sails also worked well against the blue and gold body. They provided a visual contrast without introducing another strong colour into the design. Removing the skull decoration helped achieve the simpler finish I wanted, although it also meant removing one of the original model's decorative features. An important decision was retaining the original structure of the ship. I felt that its existing shape and arrangement of components already provided a suitable foundation for the project. Completely redesigning it would have taken my work away from its main purpose, which was to customise the existing asset. Overall, I believe the final appearance achieved most of my original design goals. The ship retained its original structure but had a more consistent colour scheme and a cleaner sail design. If I were to develop it further, I would consider adding additional decorative details to certain areas without making the design look overcrowded.

One of the main difficulties I experienced while using Mecabricks was selecting individual bricks within the ship. When I first tried to select a specific brick, the entire ship would get selected instead. This made it harder to customise certain parts because I could not immediately select the exact piece I wanted to change. To work around this, I had to identify each individual brick and make changes to them one at a time. Although this took longer than I originally expected, it allowed me to continue with the customisation without affecting other parts of the ship. I had to be patient and pay attention to which bricks I was selecting, especially when working on smaller sections. This challenge taught me that editing an existing 3D model is not always straightforward. Even when I knew what changes I wanted to make, I still had to understand how the software worked before I could achieve the result I was looking for. I also realised how important it is to be careful when making changes to a model that contains many individual pieces. If I were to do this task again, I would spend more time exploring Mecabricks before starting the design. In particular, I would learn more about the selection and editing features so that I could work more quickly and confidently. Overall, although this was frustrating at times, it helped me become more patient and improved my ability to deal with problems while working on a 3D project.

In terms of the animations, the first task I completed was creating the Lego figure in Mecabricks, which would later be added to the overall project model. For my contribution, I created the ship and worked on animating its sails to make it look as though the ship was sailing with the captain on board. As shown in the screenshots below, the animation demonstrates the sails moving from left to right. They start in their original position facing forwards, then rotate towards the left, return to the middle position and finally turn towards the right. This movement is repeated to give the impression that the sails are moving naturally as the ship sails. I also made sure that both the front and back sails were animated together so that their movements were synchronised. This helped make the overall animation look more consistent and realistic, rather than having each sail move separately.



My role in Group 7 was to focus on the design and customisation of the Treasure Planet-inspired ship. Another group member was responsible for finding and importing the original ship model, while I was responsible for making changes to its appearance using Mecabricks. The main changes I made included introducing a royal blue and gold colour scheme, changing the front deck from black to blue and gold, and removing the black skull decoration from one of the sails to give it a plain white finish. These were my own design decisions, and I made them to help the ship match the appearance I had planned for our project. Alongside designing the ship, I also contributed an initial idea for the main menu interface by creating a rough sketch of how it could be laid out. I planned for the menu to have three main buttons: Enter, Help and Exit. The Enter button would allow users to move from the main menu to the ship, the Help button would display a small pop-up explaining the project and what the user can expect, and the Exit button would allow users to close the project. I wanted the menu to be simple and easy to navigate so that users could understand the available options without it being too complicated. Although this was only an initial sketch, it helped give the group an idea of how users would interact with the project before reaching the ship. Once I had finished customising the ship, I uploaded my work to my individual GitHub folder so that my contribution was saved and organised within the group's coursework repository. I then handed the completed ship model to the group member responsible for implementing it into Unity. I think this was an important part of my contribution because it meant that my work could be used in the next stage of the project. Working on this project also helped me understand how important teamwork is when creating a 3D product. Each group member had different responsibilities, but our individual contributions all supported the same final outcome. It also gave me the opportunity to work independently, take responsibility for my own tasks and contribute something creative to the group's project. Overall, I believe I contributed positively to Group 7 by improving the appearance of the original ship and preparing my completed design for the next stage of development.